

# Build the Thara network

*Day 3 - Afternoon*

Running scenario: Thara Hospitals patient portal pilot

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CONCEPT 1



# Build order matters

15 min

## Each piece needs the one before it



1. The VPC: everything else is created inside it
2. The subnets: slices of the VPC's range
3. The internet gateway, attached: a route can only point at an attached gateway
4. The route tables, then their associations
5. The instances, last

**A mis-ordered build is the commonest failure in this lab. The symptom is an SSH timeout.**

# Names, tags, and one retirement

- Names come from Thara's naming table. Type them exactly
- Every resource today: `Module=CC, Day=3, Owner=<student id>`
- Step 8 launches a new `thara-web-1` from your image
- Step 10 terminates the Day 2 server: the image is proven

## COMMON MISTAKE

*A public IP alone makes an instance reachable.*

**It needs the route too. The break-it step shows it.**

# Build the Thara network

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You build:

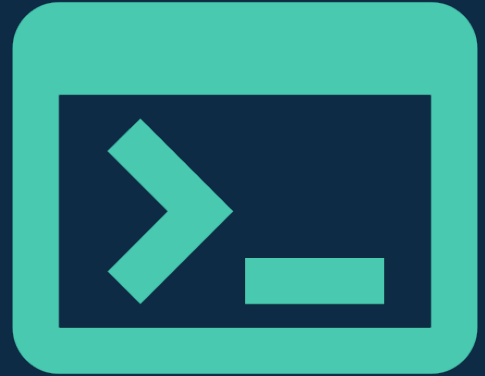
4

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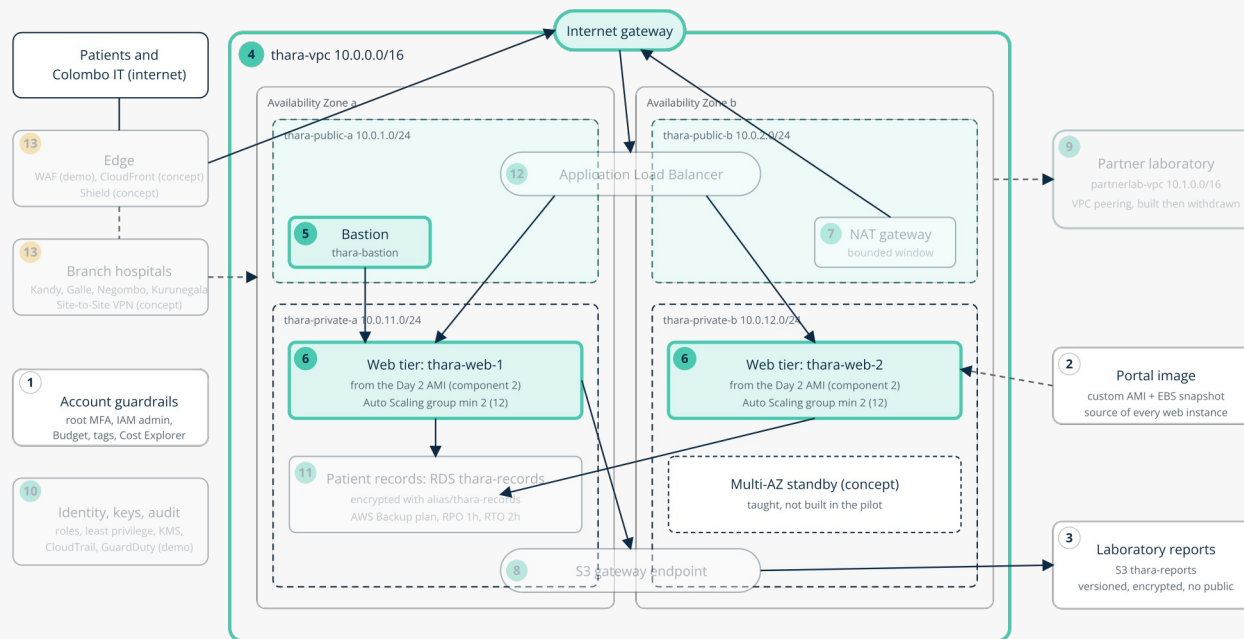
6

Estimated credit use: about USD 0.40

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# The Thara architecture



## TODAY

- 4** Thara VPC
- 5** Bastion for Colombo IT
- 6** Portal web tier

## KEY

- Built today
- Built earlier
- Not yet built

SEGMENT 1 - 45 MIN



# VPC, subnets, IGW, route tables

Cost: nil

# Check, then create the VPC



1

Confirm the region is `ap-south-1` and that you are the admin user, not root.

*Expected: Mumbai, and `admin-<student id>`, in the top bar.*

2

Create VPC, VPC only: `thara-vpc`, 10.0.0.0/16, the three tags. Then Actions, Edit VPC settings, Enable DNS hostnames.

*Expected: DNS hostnames: Enabled. A main route table and a default network ACL arrived by themselves.*

# Four subnets in two zones

3

Create `thara-public-a` 10.0.1.0/24 and `thara-private-a` 10.0.11.0/24 in zone a; `thara-public-b` 10.0.2.0/24 and `thara-private-b` 10.0.12.0/24 in zone b.

*Expected: Four subnets, two in each zone.*

**Enable auto-assign public IPv4 address on the two public subnets only. All four are still private.**

# Gateway, route tables, associations



4

Create an internet gateway with the three tags. Actions, Attach to VPC, `thara-vpc`.

*Expected: State Attached.*

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`thara-public-rt`: add 0.0.0.0/0 to the gateway; associate both public subnets. `thara-private-rt`: no route added; associate both private subnets.

*Expected: Two routes in the public table, one in the private table, two associations each.*

# Two security groups



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`thara-bastion-sg`: SSH from My IP. `thara-web-sg`: HTTP from Anywhere-IPv4, and SSH with `thara-bastion-sg` chosen as the source.

*Expected: The SSH source in `thara-web-sg` begins `sg-`, not an address.*

**A group as a source means: whatever wears that group, whatever its address.**

SEGMENT 2 - 35 MIN



# Bastion and private web instance

Cost: USD 0.022 per hour for two, plus USD 0.005 for IPv4

# Launch the bastion and connect

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Launch `thara-bastion`: Amazon Linux, t3.micro, key pair `thara-key`, subnet `thara-public-a`, existing group `thara-bastion-sg`, the three tags.

*Expected: A prompt of the form `ec2-user@ip-10-0-1-...`*

```
# in your own terminal, in the folder holding the key
$ ssh -i thara-key.pem ec2-user@<bastion public IP>
```

**A timeout here is a build-order fault. Do not launch another instance.**

# Launch the web server from your image

---

8

Launch `thara-web-1` from My AMIs, `thara-portal-v1`: subnet `thara-private-a`, Auto-assign public IP disabled, existing group `thara-web-sg`, tags with Day=3.

*Expected: A private address 10.0.11.x and no public address.*

**No user data. The web server is already in the image.**

# Reach it through the bastion

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From the bastion, SSH to the private address of `thara-web-1`. Use agent forwarding; the lab sheet has the fallback.

*Expected: The pilot page is served. curl to the internet times out.*

```
$ ssh-add thara-key.pem
$ ssh -A ec2-user@<bastion public IP>
$ ssh ec2-user@<web private IP>
$ curl http://localhost
$ curl -m 10 https://aws.amazon.com
```

# Retire the Day 2 server



10

Two servers are named `thara-web-1`. Terminate the stopped one tagged Day=2 in the default VPC, then delete its launch-wizard security group.

*Expected: One `thara-web-1` remains, in `thara-vpc`.*

**Check the Day tag before you terminate. Leave the default VPC itself alone.**

SEGMENT 3 - 25 MIN



# Security group experiments

Cost: nil

# Experiment: an address is not a group

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In `thara-web-sg`, change the SSH source from `thara-bastion-sg` to My IP. From the bastion, try SSH to `thara-web-1`. Say why. Revert.

*Expected: Connection timed out, then a connection again after the revert.*

```
# on the bastion
```

```
$ ssh -o ConnectTimeout=10 ec2-user@<web private IP>
```

SEGMENT 4 - 25 MIN



# NACL experiment

Cost: nil

# Experiment: a firewall that forgets

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Create `thara-strict-nacl`: inbound 80 and 22, outbound 80 and 443. Associate it with `thara-public-a`. SSH to the bastion. Then add outbound 1024-65535.

*Expected: A timeout, then a connection once the return rule exists.*

**Revert:** remove `thara-public-a` from its subnet associations.

SEGMENT 5 - 20 MIN



# Flow Logs

Cost: negligible

# Record the traffic, find two rejections

**13** Create log group `thara-flow-logs`. On `thara-vpc`: Actions, Create flow log, filter All, to CloudWatch Logs, new service role. Make three attempts that must fail.

*Expected: After five to ten minutes, two REJECT records for port 3389.*

```
# 1 and 2 on your own computer, 3 on the bastion
$ ssh -o ConnectTimeout=5 ec2-user@<web private IP>
$ ssh -p 3389 -o ConnectTimeout=5 ec2-user@<bastion public IP>
$ ssh -p 3389 -o ConnectTimeout=5 ec2-user@<web private IP>
```

**Attempt 1 leaves no record. It never reached the VPC.**

SEGMENT 6 - 15 MIN



# Ledger, teardown, evidence

Cost: nil

# Prove it, then break it



## PROVE IT

- From the bastion, curl to 10.0.11.x returns the pilot page
- From `thara-web-1`, curl to the internet times out
- Two REJECT records in `thara-flow-logs`, annotated

## BREAK IT

- Remove the 0.0.0.0/0 route from `thara-public-rt`
- SSH to the bastion times out, yet it still has a public IP
- Find the evidence, say why, restore the route

# Tear down and record



## TEARDOWN

- Stop `thara-bastion` and `thara-web-1`; do not terminate
- Delete `thara-strict-nacl` once it has no subnet
- Keep the VPC, route tables, security groups and Flow Log

## COST

**Estimated credit use: about USD 0.40**

## FOR YOUR EVIDENCE PACK

- Hand-drawn or tool-drawn diagram of `thara-vpc` with subnets, route tables and both instances
- Screenshot of `thara-public-rt` and `thara-private-rt` side by side
- The two annotated REJECT records
- Ledger row

# Common mistakes



## COMMON MISTAKE

*The private instance cannot be reached at all.*

**It can, from inside the VPC.**



## COMMON MISTAKE

*A public IP alone makes an instance reachable.*

**It needs the route too.**

## Before the next session



### READ

#### Amazon VPC User Guide

NAT gateways, VPC peering basics, Gateway endpoints.

### SELF-CHECK

#### CLF-C02 bank

Domain 3 networking set B, 15 items.

*PQ3 is due.*

### TASK

#### One paragraph

Why is VPC peering non-transitive, and what does that force a group with five sites to do?

*Bring it to Day 4.*

# Questions & Discussion



*Next: a way out for the private tier, and a partner network.*



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